# Critical Path Template · Expert version

A critical path template helps you run the critical path method: list activities, durations and predecessors, calculate early and late dates and float, and find the chain of activities that sets the project's end date.

## How to use this template

1. List activities, durations (days) and predecessors.
2. Forward pass: ES = the largest EF of all predecessors (0 for the first); EF = ES + duration.
3. Backward pass: LF of the last activity = its EF; LS = LF − duration; work backwards using the smallest LS of successors.
4. Total float = LS − ES; activities with zero float are critical.
5. Focus monitoring on the critical path; see forward and backward pass.

## Tips

- Re-run the calculation when durations change.
- Watch near-critical paths with little float.
- Use schedule compression only on critical activities.
- Software calculates CPM automatically for large projects.

## CPM · Enter ES from the forward pass and LS from the backward pass; float and critical flag calculate

| Activity | Description | Duration | Predecessors | ES | EF | LS | LF | Total float | Critical? |
|---|---|---|---|---|---|---|---|---|---|
| A | Requirements | 4 | | 0 | 4 | 0 | 4 | 0 | Yes |
| B | Design | 3 | A | 4 | 7 | 4 | 7 | 0 | Yes |
| C | Procure hardware | 2 | A | 4 | 6 | 8 | 10 | 4 | No |
| D | Build | 5 | B | 7 | 12 | 7 | 12 | 0 | Yes |
| E | Install hardware | 2 | C | 6 | 8 | 10 | 12 | 4 | No |
| F | Test and launch | 2 | D, E | 12 | 14 | 12 | 14 | 0 | Yes |
| | | | | | | | | | |
| | | | | | | | | | |
| | | | | | | | | | |
| | | | | | | | | | |
| | | | | | | | | | |
| | | | | | | | | | |
| | | | | | | | | | |
| | | | | | | | | | |